

module-09-subtraction-nikhilam - Darshana (60-min workshop)

IKS Curriculum - middle band

Module 9 — Subtraction by Nikhilam (Darśana)

Sanskrit: *Nikhilam Navataścaramaṁ Daśataḥ* · **Band:** Middle (6-8) · **Duration:** 60 min · **Path:** Darśana (Workshop)

“All from 9, last from 10” — a complement method for subtraction.

Darśana note. This workshop condenses Module 9’s full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to [Adhyayana](#) (the Full Course path).

Opening hook · 5 min

Project: $1000 - 437 = ?$ Long subtraction has many borrows. “Or — *Nikhilam*: all from 9, last from 10. $\rightarrow 999 - 437 = 562$, then $+1 = 563$. Today’s lesson.”

Core idea · 15 min

Nikhilam method for subtraction from powers of 10:

For $10^n - N$: 1. Subtract each digit of N from 9 EXCEPT the last digit. 2. Subtract the LAST digit from 10. 3. Concatenate.

Example: $1000 - 437$. $-4 \rightarrow 9 - 4 = 5$ - $3 \rightarrow 9 - 3 = 6$ - $7 \rightarrow 10 - 7 = 3$ - Answer: 563.

Algebraic identity: $10^n - N = (10^n - 1) - N + 1$ = “all 9s minus N , plus 1.” The procedure encodes this.

Demonstration · 10 min

Live computation. Compute $10000 - 4783$ by both methods. Time it. Discuss why Nikhilam is faster (no borrowing cascade).

Source moment · 5 min

Paraphrase Tirthaji 1965, Sūtra 2: “*Nikhilam Navataścaramam Daśataḥ — all from 9 and the last from 10. The complement method.*”

Hands-on activity · 15 min

Speed race. 20 problems mixing subtraction from 100, 1000, 10000. Each student records their time + accuracy.

Reflection + take-home · 10 min

“*What surprised you most about the connection between Nikhilam and computers?*”

Go deeper into Adhyayana

The 10 days you’re skipping if you only do this workshop:

- **Day 01** — The sūtra — all from 9, last from 10
- **Day 02** — Complement to 1000
- **Day 03** — Subtract from a power of 10
- **Day 04** — General subtraction
- **Day 05** — The algebra — why it works
- **Day 06** — Bar model proof
- **Day 07** — Two’s complement in binary
- **Day 08** — Speed practice
- **Day 09** — Mixed practice
- **Day 10** — Final assessment + computer-subtraction poster

→ [Open the Adhyayana track for this module](#) to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.